Project-1: Online Retail Sales Database Design

1. Introduction

In today's digital economy, e-commerce platforms rely heavily on well-structured databases to manage customers, products, orders, and payments efficiently. A poorly designed database can lead to redundancy, inconsistency, and performance issues. Therefore, designing a normalized and scalable relational database is a critical step in building any online retail system.

This project focuses on designing and implementing a normalized SQL database schema for an Online Retail Sales platform. The goal was not only to create tables but to design a structure that reflects real-world business logic, maintains data integrity, and supports reporting and analytics requirements. The project emphasizes normalization up to Third Normal Form (3NF), proper use of constraints, and optimized relational modelling.

2. Abstract

The objective of this project was to design a structured relational database for an e-commerce platform using SQL. The database includes essential entities such as Customers, Products, Categories, Orders, Order Items, Payments, and Inventory.

A detailed Entity-Relationship (ER) diagram was created to model relationships between entities. Special attention was given to handling many-to-many relationships using a junction table (Order_Items) and preserving historical data such as product prices at the time of purchase.

The schema was normalized up to 3NF to eliminate redundancy and ensure consistency. Primary keys, foreign keys, unique constraints, check constraints, and composite keys were implemented to maintain referential integrity and data validation. Sample data was inserted to simulate real-world operations, and advanced JOIN queries were written to generate meaningful sales reports.

This project demonstrates practical database design skills aligned with industry standards and real-world e-commerce systems.

3. Tools Used

- **MySQL** – For database creation and execution of SQL scripts
- **dbdiagram.io** – For designing and visualizing the ER diagram
- **SQL (DDL & DML)** – For schema creation, constraints implementation, and data manipulation

4. Steps Involved in Building the Project

Step 1: Requirement Analysis

Step 2: Entity-Relationship Design

Step 3: Normalization

The schema was normalized up to Third Normal Form (3NF):

- Ensured atomic values (1NF)
- Removed partial dependencies (2NF)
- Eliminated transitive dependencies (3NF)

Step 4: Schema Implementation (DDL Scripts)

Step 5: Data Population

Step 6: Reporting Queries and Views

5. Conclusion

This project successfully demonstrates the design and implementation of a normalized relational database for an online retail system. The schema was built using industry-level design principles, ensuring scalability, data integrity, and efficient querying.

By applying normalization techniques, enforcing constraints, and handling complex relationships properly, the database structure aligns with real-world e-commerce requirements. The inclusion of reporting queries and views further enhances its practical applicability.

Overall, this project strengthened understanding of relational modelling, normalization, constraint management, and advanced SQL querying. It reflects a production-oriented approach rather than a basic academic implementation, preparing the foundation for building scalable backend systems in real-world applications.